QP CODE: 25900016

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025
SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCBTS100 - BIOLOGICAL FOUNDATIONS FOR HEALTH AND WELLNESS
(2024 Admission onwards)

Duration: 1 Hours **Maximum Marks: 35**

Part A
Multiple Choice Questions
Answer **All** questions. Each question carries **1** mark.

1.What is the term for the different forms of a single gene? [K] [CO3]
Answer: **a) Allele**
2.What is the function of the ovary in a flowering plant? [U] [CO3]
Answer: **b) To produce ovules**
3.What is the primary female reproductive hormone? [K] [CO2]
Answer: a) **Estrogen**
4. What is the main role of GABA in the brain? [U] [CO5]
Answer: **d) Inhibitory neuro transmission**
5.What connects muscles to bones? [K] [CO2]
Answer: **b) Tendons**

(5 x 1 = 5 Marks)

Part B

Short Answer Questions.
Answer any five questions. Each question carries 2 marks.

6. Give two functions of carbohydrates. [U] [CO1]

Primary functions of carbohydrates in the body are energy production, energy storage, building macromolecules, helping in lipid metabolism.

7. How would damage to the pituitary gland affect other endocrine glands? [A] [CO5]

Pituitary gland produces hormones that stimulate other glands to release their own hormones. Damage to pituitary gland can disrupt this regulation, leading to imbalances in various bodily functions.

8.What role does protein play in muscle repair and growth? [U] [CO4]

Protein is crucial for muscle repair and growth as it provides amino acids needed to rebuild and strengthen muscle tissue damaged during exercise. When muscles are worked, they experience microscopic tears, and protein intake supports the body's ability to repair these tears and stimulate new muscle protein synthesis, leading to growth and increased strength.

9.What are proteins? Give examples. [K] [CO1]

Proteins are organic molecules composed of amino acids linked together by peptide bonds. They play a crucial role in various biological functions, that includes structural support, enzyme activity, and immune function. Examples of proteins include collagen in skin, keratin in hair and nails, haemoglobin in RBC, and enzymes such as amylase in saliva.

10.What is excretory system? [K] [CO2]

The excretory system in humans is a network of organs responsible for removing waste products and excess substances from the body, helping to maintain a balanced internal environment and preventing damage. It primarily includes the kidneys, ureters, bladder, and urethra and also involves the liver, lungs and skin.

11.Which level includes all the living and non-living components in an area? [K] [CO1]

The ecosystem level of ecological organization includes all living (biotic) and non-living (abiotic) components in a particular area, interacting as a system. Biotic factors are the living components, such as plants, animals, and microorganisms, that interact within the ecosystem. Abiotic factors are the non-living components, such as sunlight, water, temperature, soil, and nutrients, that influence the living organisms in the ecosystem.

12.Explain the difference between innate and adaptive immunity. [U] [CO5]

Innate immunity is the body's immediate, non-specific defence against pathogens, while adaptive immunity is a more specialized and long-lasting response that develops after exposure to a specific pathogen. Innate Immunity does not target a specific type of pathogen but responds to a broad range of invaders. It is present at birth providing a rapid response. It includes skin, mucous membranes and phagocytes

Adaptive Immunity recognizes and targets specific pathogens. It develops over time as the body is exposed to pathogens or vaccines. It involves B cells (which produce antibodies) and T cells (which directly attack infected cells).

(5 x 2 = 10 Marks)

Part C

Short Essay Questions
Answer any **five** questions. Each question carries **4** marks.

13.Explain the factors enhancing physical and mental health. [K] [CO1]

factors significantly impact both physical and mental health. These include lifestyle choices like regular exercise, a balanced diet, sufficient sleep, and avoiding harmful substances. Furthermore, strong social connections, positive relationships, a sense of purpose, and financial stability play a crucial role in overall well-being.

1)Balanced Diet: Eating a nutritious diet provides the body with essential nutrients, supporting physical health and potentially improving mental well-being.

2)Regular Exercise: Physical activity, in any form, is beneficial for both physical and mental health. It can improve mood, reduce stress, and enhance self-esteem. Even short bursts of activity can make a difference.

3)Sufficient Sleep: Adequate sleep is crucial for both physical and mental restoration. It helps the body repair tissues, boosts the immune system, and allows the mind to rest.

4)Avoidance of Harmful Substances: Limiting or avoiding alcohol and drug use is essential for both physical and mental health.

5)Strong Social Connections: Building and maintaining positive relationships with family and friends provides emotional support and reduces stress.

14.Explain the different chronic diseases due to junk food consumption. [U] [CO4]

Junk food consumption is linked to several chronic diseases due to its high calorie, fat, and sugar content, while lacking essential nutrients. These include obesity, heart disease, type 2 diabetes, and certain cancers. Additionally, junk food can contribute to digestive issues and mental health problems

Specific Chronic Diseases:

1)Obesity: Junk food is often high in calories and fat, leading to weight gain and obesity, a major risk factor for many chronic diseases.

2) Heart Disease: The high saturated fat and sodium content in junk food can raise cholesterol levels and blood pressure, increasing the risk of heart disease.

3)Type 2 Diabetes: The high sugar and processed carbohydrates in junk food can lead to insulin resistance and an increased risk of developing type 2 diabetes.

4)Certain Cancers: Studies suggest that diets high in junk food may be linked to an increased risk of certain cancers, including colon, rectal, and oesophageal cancer.

5)Digestive Issues: Junk food can contribute to digestive problems like constipation, diarrhoea, and acid reflux.

6)Mental Health Problems:Some studies suggest a link between high junk food consumption and an increased risk of depression and other mental health issues.

15.Explain gene expression. [K] [CO3]

Gene expression is the process by which the information encoded in a gene is used to create a functional product called protein. This process involves two main steps: transcription, where DNA is copied into RNA, and translation, where RNA is used to synthesize a protein.

1. Transcription:

- The genetic information in DNA is transcribed into a mRNA molecule.
- This process occurs in the nucleus of eukaryotic cells.
- Specific enzymes RNA polymerase, and other proteins, are involved in this process.
- The resulting mRNA molecule carries the genetic code from the DNA

2. Translation:

- The mRNA molecule travels to the cytoplasm, where it interacts with ribosomes and t-RNA molecules.
- Ribosomes read the mRNA sequence in codons
- tRNA molecules bring the corresponding amino acids to the ribosome, which then assembles them into a polypeptide chain.
- This polypeptide chain folds into a functional protein.

16.How does neurotransmitter imbalance contribute to mental health disorders like depression and anxiety? [An] [CO5]

Neurotransmitter imbalances, characterized by either too much or too little of specific brain chemicals, leads to mental health disorders like depression and anxiety. These imbalances

can disrupt normal brain function, leading to changes in mood, behaviour and emotional regulation. Neurotransmitters and their Roles:

- Serotonin: Plays a crucial role in mood regulation, sleep, appetite, and pain perception. Imbalances, particularly low levels, are linked to depression and anxiety.
- Norepinephrine: Involved in alertness, attention, and the fight or flight response. Low levels can contribute to depression, while high levels may lead to anxiety and hyperactivity.
- Dopamine: Associated with motivation, pleasure, and reward. Imbalances can lead to symptoms like apathy, loss of interest, and difficulty experiencing pleasure, which can contribute to depression and other mental health conditions.
- GABA: A major inhibitory neurotransmitter, helps to calm the brain. Low levels of GABA can contribute to anxiety disorders.

How Imbalances Contribute to Mental Health Disorders:

1. Disrupted Communication:Neurotransmitters act as chemical messengers, transmitting signals between neurons. Imbalances disrupt this communication process, potentially leading to abnormal brain activity.

   2. Changes in Brain Circuits: Neurotransmitters influence the activity of specific brain circuits involved in mood, emotion, and cognition. Imbalances can disrupt these circuits, leading to symptoms of depression and anxiety.

   3. Altered Responses to Stress: Imbalances in neurotransmitters can affect how the brain responds to stress, leading to increased anxiety and depression.

   4. Impact on Mood and Behaviour:Neurotransmitter imbalances can directly influence mood, leading to feelings of sadness, hopelessness, irritability, and anxiety. They can also affect behaviour, such as appetite, sleep, and energy levels.

17.Name the organs of digestive system and their functions. [K] [CO2]

The organs of digestive system and their functions are the following: -

- Mouth: Initiates digestion by mechanical breakdown (chewing) and chemical digestion (saliva).
- Pharynx: A passageway for food and air.
- Oesophagus: Transports food from the pharynx to the stomach by peristalsis.
- Stomach: Stores food, mixes it with gastric juices (acids and enzymes), and begins the breakdown of proteins.
- Small Intestine: Completes digestion and absorbs most nutrients.
- Large Intestine: Absorbs water and electrolytes, forms faeces, and stores waste before elimination.
- Rectum: Stores faeces.
- Anus: Eliminates waste from the body.

Accessory Organs:

- Salivary Glands: Produce saliva, which contains amylase that breakdown carbohydrates.
- Liver: Produces bile, which aids in the digestion of fats.
- Gallbladder: Stores and concentrates bile.

- Pancreas: Produces enzymes that digest carbohydrates, proteins, and fats, as well as hormones that regulate blood sugar.

18. What is circulatory system? Why is it important? [K] [CO2]

The circulatory system is a vital network responsible for transporting blood throughout the body. It is composed of the heart, blood and blood vessels that include arteries, veins, capillaries.

Importance of circulatory system:

- Oxygen and Nutrient Delivery: The circulatory system carries oxygen from the lungs and nutrients from the digestive system to all parts of the body. This is crucial for cellular function and energy production.
- Waste Removal: The circulatory system transports waste products, like carbon dioxide, from cells to the lungs for exhalation and other organs for elimination.
- Hormone Transport: The circulatory system also carries hormones, which are chemical messengers that regulate various bodily functions.
- Immune Function: The circulatory system helps transport immune cells to fight off infections and diseases.
- Temperature Regulation: The circulatory system plays a role in regulating body temperature by distributing heat to different parts of the body.

19. What are the preventive measures for maintaining a healthy respiratory system? [K] [CO2]

For maintaining a healthy respiratory system, prioritize preventing infections, avoiding harmful exposures, and adopting a healthy lifestyle.

Avoiding Harmful Exposures:

- Avoiding Second hand Smoke: Exposure to second hand smoke is harmful and has to be avoided.
- Minimizing Air Pollution Exposure: Limiting the time outdoors during periods of high air pollution and consider wearing a mask if necessary.
- Improving indoor Air Quality: Ventilate the home regularly and use air purifiers.

Adopting a Healthy Lifestyle:

- Regular exercise: Physical activity strengthens the lungs and improves overall respiratory function.
- Practicing Deep Breathing: Deep breathing exercises can help strengthen the respiratory muscles and improve lung capacity.
- Eating a Healthy Diet: A balanced diet rich in fruits, vegetables, and whole grains provides essential nutrients for respiratory health.
- Managing Stress: Stress can affect breathing, so finding healthy ways to manage stress, such as exercise, meditation, or spending time in nature can be beneficial.

(5 x 4 = 20 Marks)